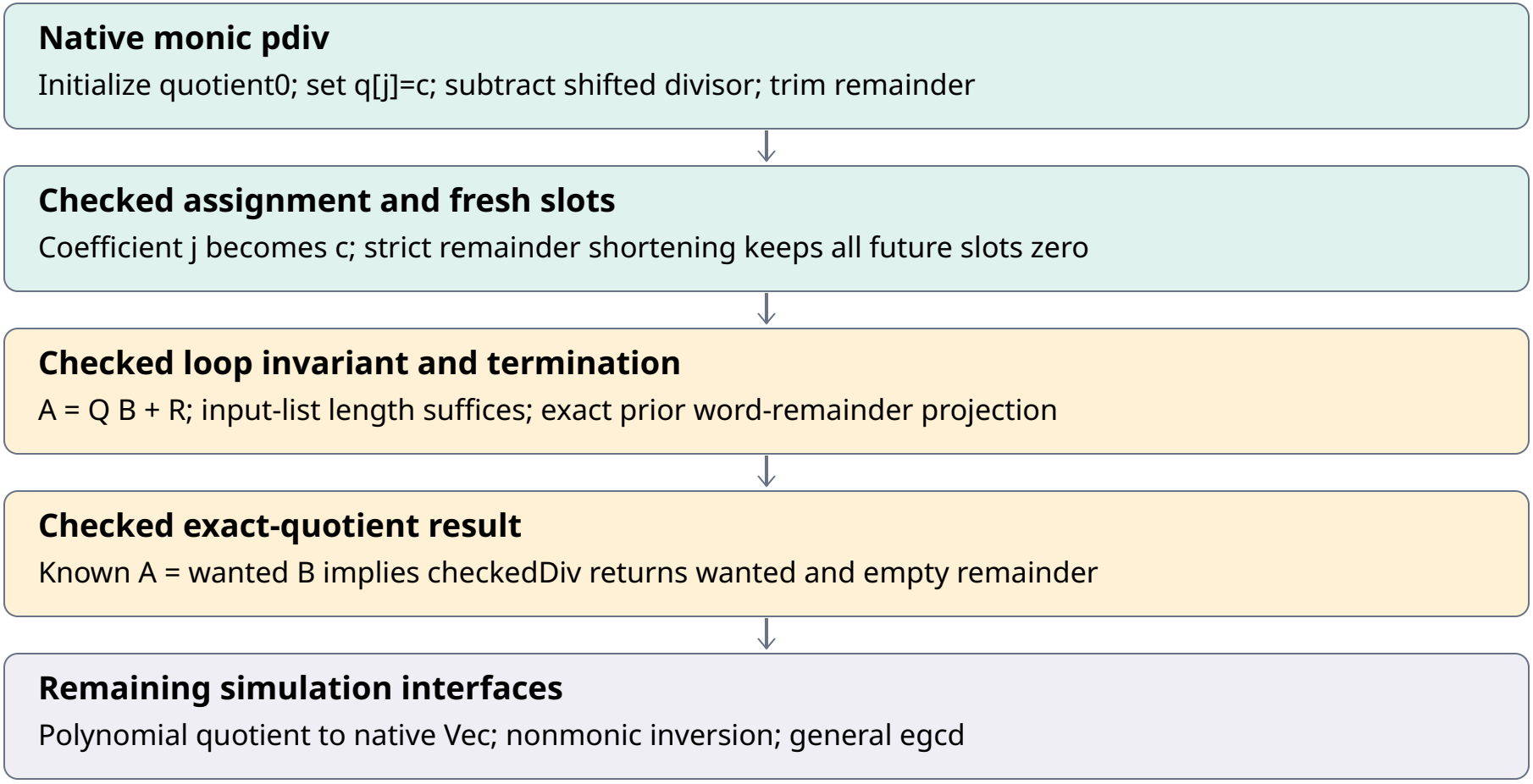Monic division: quotient accumulation correspondence